Midway Report: Agent Hospital 2.0

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Abstract

Improving healthcare systems by assisting doctors in their daily tasks is a goal that can be significantly advanced through the use of **Large Language Models (LLMs)**. One of the recent developments is the **Simulacrum-based Evolutionary Agent Learning (SEAL)** framework introduced in the **Agent Hospital** project (Li et al., 2025). Our project aims to reimplement a simplified version of the SEAL model to investigate how reducing model complexity affects its performance. This translates to constructing a Doctor agent through SEAL. We focus exclusively on **doctor-patient interactions** and use alternative data sources to build training material. Our methodology includes two major steps that can be asynchronous: generating fictional patients and having the doctor agent evolve by interacting with them. We have already started generating diverse patients to simulate various medical scenarios, aiding the doctor agent in disease prediction. And we have also built a benchmark for real-life evaluation of doctor agents. That corresponds to testing the agent on the MedQA dataset. Our benchmark consists of a PubMedBERT model and a FLAN-T5-base model that reach 0.3357 and 0.2461 of best validation accuracy respectively. From this, we have concluded that pre-training on medical data is essential to perform on the MedQA test.

We believe that our version Agent Hospital could serve as a lightweight assistant for doctors by helping filter out common conditions based on frequently reported symptoms. This could help reduce the initial workload in clinical settings and improve efficiency during early consultations. Although our results are preliminary, this mini-project serves as a proof of concept that offers valuable insights into building accessible and effective AI-powered doctor agents, particularly in resource-constrained settings.

1 Introduction

Our goal of analyzing the SEAL method using small scale AI for the implementation of a doctor agent capable of helping actual doctors can be split in two tasks. The first is the generation of fictional patients, which will then be used as data for the second phase. Although domain expertise remains important for interpreting complex medical content, we effectively started generating patients using **MedlinePlus**[1] data. The second task is training through simulation a doctor agent with the generated data. This task has not yet been initiated but we have built benchmarks for the real-life evaluation of doctor agents. Indeed, we have fine-tuned two LLMs on the **MedQA dataset**[2] (Jin et al., 2020) test which consists of multiple choice questions that require medical expertise. The models used were **PubMedBERT**[3] (GU et al., 2021) and **FLAN-T5-base**[4]. With some hyper-parameters optimization, they respectively achieved validation accuracies of 0.3357 and 0.2461. We suspect this difference in performance is due to the lack of medical terms knowledge by the FLAN-T5-base since it was not pre-trained specifically on medical data like PubMedBERT.

2 Related work

Agent Hospital: A Simulacrum of Hospital with Evolvable Medical Agents (Li et al., 2025) is the very foundation of our project. However, it is very recent (January 17, 2025) and its source code is not available. Additionally, the dataset used, **Baidu Health Encyclopedia**[5], is entirely in Chinese, lim-

[1] **MedlinePlus**:https://medlineplus.gov
[2] **MedQA dataset**:https://github.com/jind11/MedQA
[3] **PubMedBERT**:https://arxiv.org/pdf/2007.15779
[4] **FLAN-T5-base**:https://huggingface.co/google/flan-t5-base
[5] **Baidu Health Encyclopedia**: https://jiankang.baidu.com/widescreen/entitylist

iting direct reuse. As a result, we developed our approach from scratch, leveraging the capabilities of LLMs and integrating relevant course concepts. We identified a suitable alternative in **MedlinePlus**, a public resource from the U.S. National Library of Medicine, closely aligned with Baidu's dataset and accessible via API. Another limitation of this paper is that it does not explicitly explain how the learning of doctor agents is implemented during the simulation. Consequently, we will use fine-tuning on a classification task to improve our doctor agent since this method is known to work efficiently.

BERTScore: Evaluating Text Generation with BERT (Zhang et al., 2020) presents a promising approach for evaluating our generated patient agents. It computes precision, recall, and F1 scores by comparing contextual embeddings at the token level, without relying solely on sentence-level representations. Its strengths include fine-grained evaluation, open-source [6] availability, and easy installation via `pip install bert-score`. The original paper primarily addresses machine translation and image captioning tasks, lacking experiments on other types of text generation, such as summarization or open-ended generation (as in our case), which limits its demonstrated flexibility. To address this, we propose a **hybrid method combining BERTScore** with cosine similarity from aggregated sentence embeddings, offering a more comprehensive evaluation.

3 Methods

3.1 Retrieving Datasets for Patient Agent Generation

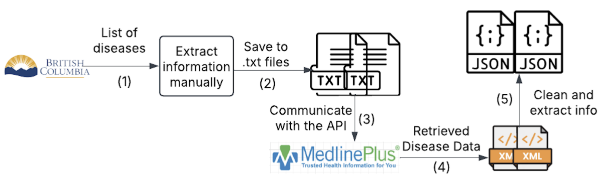

Figure 1: Disease Data Extraction and Processing Pipeline

The process begins by retrieving a list of diseases associated with each medical department from the **Government of British Columbia website**[7], as illustrated in **Step 1** of Figure 1. This list, originally intended for doctor billing purposes, contains diagnostic codes and administrative terms that are not directly suitable for our task. Therefore, manual adjustments with guided by a domain expert were performed to extract meaningful disease names. As part of **Step 2**, the cleaned disease list is saved as .txt files and then used in **Step 3** to send requests to the **MedlinePlus API** [8]. The retrieved disease information, returned in XML format at **Step 4**, is then cleaned, structured, and saved as JSON files at **Step 5**.

3.2 Agent Patient Generation

The synthetic patients are generated using the **LLaMA 2 language model** via the *Ollama interface*. Prompts are sent to a locally hosted Ollama server running the LLaMA 2 model. For each disease (retrieved from Section 3.1), which includes a *title, diagnostic cause*, and *a textual summary*, the system constructs a structured prompt that instructs the model to generate a synthetic patient profile in *JSON format*. Each profile contains the following fields: *Name, Age, Gender, Symptoms, Symptom duration, Recent travel history, Medical history, Treatment plan, Correct disease, Diagnosis cause*, and *Mixed diseases (a shuffled list of related conditions)*. The generated output is stored in a *.txt files* for further processing.

3.3 Modification of Patient Agent Information

To enhance the robustness and adaptability of the doctor agent, we introduce controlled noise into the symptom data. Leveraging the synthetic patient profiles generated in Section 3.2, we prompt the model to randomly alter patient agent information by selectively removing relevant symptoms or slightly modifying existing symptoms within their symptom lists.

3.4 Doctor Agents baseline

To establish an initial performance baseline for the doctor agent, we have conducted a benchmark by fully fine-tuning and optimizing two pretrained language models on the MedQA dataset which corresponds to the real-life evaluation of doctors. The models selected for comparison include **PubMedBERT** (a BERT variant specialized in biomedical texts) and **FLAN-T5-base**. We have chosen these models because of their high understanding capabilities but also because they can be fine-tuned in

[6] **BERTScore GitHub** https://github.com/Tiiiger/bert_score

[7] **Government of British Columbia** https://www2.gov.bc.ca/gov/content/home

[8] **Medline Plus API** https://wsearch.nlm.nih.gov/ws/query?db=healthTopics&term=CHRONICHYPOTENSION

a feasible amount of time. More details about this benchmark will be given in the Section 4.2.
To top it off, we also plan on **pre-training LLMs** on the MedQA textbooks before the fine-tuning to have stronger baselines. We will also use these models in the simulation environment to see how they compare to our medical agent for identifying diseases.

3.5 Agent Doctor Responses Evolution

In this phase, the doctor agent is exposed to synthetic patients generated in Section 3.2, where the correct diagnosis is not disclosed. An example of such a patient agent is provided in Appendix B. For each interaction, the doctor agent must select the correct diagnosis from the provided list of *Mixed_diseases*, simulating a multiple-choice question format similar to the MedQA test. Whenever the model makes an incorrect prediction, the correct answer is revealed, allowing it to update its knowledge base and learn from its mistakes. We refer to this iterative learning process as **virtual-environment training**. To assess **real-world performance**, the agent is subsequently re-evaluated on the official MedQA, a collection of multiple choice questions from professional medical board exams. These two methods of evaluation are inspired by **Agent Hospital** (Li et al., 2025). The similarity in format between the virtual-task and the real-world-task will ease the transfer between the two for the doctor agents. The goal is to outperform the baseline established in Section 3.4 on the MedQA but also in the simulation task. If time permits, we plan to implement a **mistake tracking mechanism**, which prioritizes repeated exposure to diseases the model consistently misclassifies, also known as *hard examples*. Additionally, we aim to *store these mistakes in a long-term memory module* to enable periodic revisiting, reinforcing learning through spaced repetition.

3.6 Evaluation of Patient Agent Generation

3.6.1 Expert Clinical Plausibility

This **manual step** is performed only at the beginning, as part of the midway report, to assess the model's ability to generate realistic patient profiles. Each profile is reviewed by a clinical expert to identify any false or hallucinatory information and to evaluate the overall coherence of the content.

3.6.2 Symptom Matching

We would like to apply the idea of the **hybrid evaluation method** proposed from 2, the *cosine similarity* and *BERTScore* will be automatically computed between fields: the symptoms from the full disease summary and those generated in the patient's medical profile. These scores are then compared against predefined thresholds described in Section 4.3.

4 Experiments

4.1 Datasets

We use MedQA[2] questions and answers as a baseline, along with its textbooks for both baseline models and pre-training our doctor agent, although most training will rely on synthetic data. The first 20 generated patient profiles are available online in the **Generated Patient Dataset** [9] and will be updated regularly. Curated **lists of diseases by department** are also available online[10]. Since profile generation is based entirely on MedlinePlus data, some information may be outdated. Due to the high cost of manual updates, we use the source data as-is.

4.2 Baselines

For baselines, we have already implemented two full fine-tuned models on the MedQA task. For both of them, similar data preprocessing was needed by joining the questions with their options in a specific manner. For the BERT model, a classification head was added and the T5 model generated the answer's letter with its decoder. Both models used cross entropy loss for training. In addition, we have run a grid search on the learning rate ($5\mathrm{e}^{-3}$, $5\mathrm{e}^{-4}$ and $5\mathrm{e}^{-5}$) and on the number of epochs $(3, 4, 5, 6)$ for the hyper-parameter optimization. The search was based on validation accuracy and the best results were obtained for PubMedBERT with a learning rate of $\mathbf{5e^{-4}}$ and **5 epochs of training**. The FLAN-T5-base was optimal with the same learning rate but **4 epochs** of training instead.

4.3 Evaluation Methods

Evaluation of Patient Agent Generation: At this stage, we focus on generating synthetic patient profiles. *Symptom matching* (Section 3.6.2) is planned but not yet implemented. We will start with thresholds of *cosine similarity > 0.7* and *BERTScore*

[9]**Patient Dataset** https://drive.google.com/drive/folders/1XmuahaqRpTiQx4P9Qop544XikkrT47Gp

[10]**Disease Reference** https://drive.google.com/drive/folders/1-0HNaC0DaudePO2ZbMwqwk-SirGxrRBD

F1 > 0.65, using full disease summaries as *references*. For now, *expert-assisted human judgment* (Section 3.6.1) provides initial validation based on the plausibility and coherence of the profiles.

Evaluating the doctor agents: We always use the accuracy since it reflects really well the performance in a multiple choice answering framework. Indeed, the goal of our medical agent is to have the best possible accuracy on the MedQA test but also the best possible accuracy when diagnosing patients.

4.4 Experimental Details

Since we have only worked on patients generation and baseline implementation, the experimental details have already been discussed in previous sections.

4.5 Results

4.5.1 Agent Patient Generation

We created a list of relevant heart disease terms and used it to generate synthetic patient profiles. An example of the cleaned disease list and a sample generated patient profile can be found in Appendices A and B, respectively.

4.5.2 Doctor Agent

The results we have for now are presented in table 1. From this we see that the BERT model achieves better results than the T5 and with less training time. The difference in performance is, in our opinion, caused by the difference in pre-training on medical data. That is because there is not enough training data for a model to learn medical terms during fine-tuning. On the other hand, the difference in training time might come from the fact that the T5 model has to train its decoder part as well as its encoder compared to the BERT. In addition, training these two components does not seem efficient for multiple choice answering. Therefore, we will continue our work using PubMedBERT as the foundation for future baselines and for the base of our doctor agent model.

	Validation Accuracy	Training Time (hours)
PubMedBERT	0.3357	0.8222
FLAN-T5-base	0.2461	1.7939

Table 1: Validation accuracy and training time of PubMedBERT and FLAN-T5-base fine-tuned and evaluated on MedQA. Both models have learning rates of $5e^{-4}$. The former was trained for 5 epochs and the latter for 4 epochs.

5 Future Work

- **Evaluation of Patient Agent Generation** *(April 3–5)* — Refer to Section 4.3: This step ensures the generated patient profiles are suitable for later stage of the project.
- **Modification of Patient Agent Information** *(April 6–12)* — Refer to Section 3.3: This step introduced controlled variation and noise into the patient data, enhancing the model's learning and robustness.
- **Complexify Doctor Agent Baselines** *(April 3–12* — Refer to Section 3.4: This step helps to achieve better results with and without simulation. It is also to better replicate the work of Agent Hospital (Li et al., 2025).
- **Doctor Agent Response Evolution** *(April 12–24, to be completed by April 25)* — Refer to Section 3.5: This step helps the doctor agent learn from feedback and improve diagnoses over time.

6 Task Division

Completed Task	**Assignee**
Patient Agent Generation	Nhung
Doctor Agent Baselines	Philippe

Table 2: Completed Tasks and Assignment

Task	**Assignee**
Patient Agent Generation Evaluation	Nhung
Patient Agent Modification	Nhung
Complexify Doctor Agent Baselines	Philippe
Doctor Agent Response Evolution	Philippe

Table 3: Planned Tasks and Assignment

This task division mainly serves as a reference for each team member's responsibilities. Since tasks are carried out in parallel, close collaboration is required, as progress in one area often supports another. For example, modifying patient agent information depends on feedback from the doctor agent's response and vice-versa. Tasks may need to be revisited multiple times to monitor improvements and adapt to the evolving timeline.

References

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A Appendix Manually Adjusted Diseases List

This is an exemple of this manually adjusted diseases list, the full list can be found at https://drive.google.com/file/d/1XiyoIVXp3H7Q-LZcL6XM9GTs8Zsbs6MS/view

```
PERICARDITIS
ENDOCARDITIS
MYOCARDITIS
HAEMOPERICARDIUM
CONSTRICTIVE PERICARDITIS
MITRAL VALVE DISORDERS
AORTIC VALVE DISORDERS
TRICUSPID VALVE DISORDERS
PULMONARY VALVE DISORDERS
```

Figure 2: Extracted heart disease terms

B Appendix Sample Generated Patient Profile

This is an example of a generated patient profile, where the correct disease is *Atrial Fibrillation*.

```
Full Name: John Doe
Age: 62
Gender: Male
Symptoms: Dizziness, fatigue, and a pounding heartbeat
Symptom duration: 2 weeks
Recent travel history: No recent travel
Medical history: Hypertension, hyperlipidemia, and obesity
Treatment plan: Aspirin 81mg daily, beta blocker therapy, and catheter
    ablation procedure scheduled

John is a 62-year-old male patient who has been experiencing dizziness,
    fatigue, and a pounding heartbeat for the past two weeks. He has a
    history of hypertension, hyperlipidemia, and obesity, which may
    contribute to his symptoms. He has not recently traveled anywhere and
    does not have any known allergies or sensitivities. John's family
    history is significant for cardiovascular disease, with multiple first-
    degree relatives who have been diagnosed with AFib.

Based on John's symptoms and medical history, it is likely that he is
    experiencing atrial fibrillation (AFib). The patient's recent onset of
    symptoms, along with his age and medical history, make AFib a probable
    diagnosis. To confirm the diagnosis, John will undergo further
    diagnostic testing, such as an electrocardiogram (ECG) or holter
    monitor, to assess his heart rhythm and identify any abnormalities.

In light of John's symptoms and medical history, a treatment plan has been
    developed to manage his AFib. Aspirin 81mg daily will be initiated to
    reduce the risk of stroke, while beta blocker therapy will be initiated
    to control John's heart rate and rhythm. A catheter ablation procedure
    is also scheduled to remove any damaged tissue in John's heart that may
    be contributing to his AFib. By managing John's symptoms and underlying
    conditions, the goal of treatment is to restore his normal heart rhythm
    and improve his quality of life.
Mixed diseases:
- Pulmonary Hypertension
- Anesthesia
- Atrial Fibrillation
- Bacterial Infections
- Menopause

Correct disease:
Atrial Fibrillation
Diagnosis cause:
chronic hypotension
```

Figure 3: Generated patient profile for Atrial Fibrillation